



BUMAL SATHSARA

DATA SCIENCE UNDERGRADUATE

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ABOUT ME

Final year Data Science undergraduate at SLIIT with hands-on industry experience gained through a 6 month internship at Sri Lanka Telecom (SLT), where I worked on AI engineering projects involving RAG systems, LLMs, and MCP servers. Passionate about building intelligent, data-driven solutions and always eager to take on new challenges in the AI and Data Science space.

WORK EXPERIENCE

Sep 2025 – Mar 2026 (6 months)

Data Science Intern

Sri Lanka Telecom (SLT), Head Office - Sri Lanka

- Built and deployed AI-powered applications using LLMs, RAG systems, and MCP servers in a real-world telecom environment.
- Contributed to internal data pipelines and reporting workflows within the data science team.
- Gained exposure to enterprise-scale data infrastructure and business intelligence tools.

EDUCATION

- BSc (Hons) in IT Specializing in Data Science** 2023 - Present
Sri Lanka Institute of Information Technology (SLIIT)
4th Year Undergraduate
- Higher Diploma In Information Technology** 2024
Sri Lanka Institute of Information Technology (SLIIT)
- G.C.E.(A/L) in Physical Science Stream** 2021
Rahula College, Matara
Combined Mathematics - C, Chemistry - C, Physics - S

PROJECTS

RAG-Based Circulars Chatbot

2026

- Built an end-to-end Retrieval Augmented Generation (RAG) system for intelligent querying of organizational circulars.
- Implemented OCR pipeline using Tesseract for PDF extraction, document chunking, and LLM-based metadata tagging.
- Designed hybrid search combining BAAI/bge-m3 semantic embeddings, TF-IDF lexical search, and temporal recency scoring.
- Developed two custom score fusion algorithms (Multiplicative Gating and Penalty-Scored Fusion) for optimized retrieval accuracy.
- Integrated Groq API for contextual response generation and stored data in PostgreSQL with JSONB metadata.
- Technologies used: **Python, PostgreSQL, RAG, NLP, LLM, Groq API, sentence-transformers, BAAI, Vector Embeddings, Semantic Search, TF-IDF, scikit-learn, Hybrid Search, OCR, Tesseract, pdf2image, Streamlit, Pandas, NumPy, psycpg2, JSONB, Information Retrieval, Document Processing**

Smart Fish Transportation Quality Monitoring System | IoT & Visual Analytics Dashboard

2026

- Developed an IoT-based system to monitor fish quality in real time during transportation using ESP32 sensors.
- Transmitted temperature, vibration, inbound/outbound and cargo weight data via MQTT over WiFi to a Node.js backend.
- Built an interactive React dashboard with Chart.js for logistics and quality assurance decision-making.
- Technologies used: **ESP32, IoT Sensors, MQTT, Node.js, React, Chart.js, MongoDB, REST API, Data Visualization**

Skillora | The Ultimate Social Platform for Skill Sharing

2025

- We developed a web-based social media platform using a Java (Spring Boot) based REST API for skills sharing.
- Technologies used: **React, Vite, Tailwind CSS, Spring Boot, Rest API, MongoDB, Cloudinary, OAuth2**

- Developed an AI-powered hotel review analysis system using a multi-agent architecture. The platform automates sentiment classification, scoring, topic extraction, summarization, and business recommendation generation. We implemented a Two-Stage AI workflow combining **HuggingFace transformer models and Google Gemini LLMs** to process customer feedback end-to-end. The system includes **real-time dashboards, analytics visualizations**, and **batch processing** capabilities within a **Django**-based web application.
- Technologies used: HuggingFace Transformers(**RoBERTa, BERT, BART**), **Google Gemini LLM, Django, REST APIs, PostgreSQL, Multi-Agent Systems, NLP, Machine Learning, Chart.js, Bootstrap, Python**

Data Warehousing and Business Intelligence Project

2025

- Designed a data warehouse schema using Dimensional Modeling principles. Then I developed the data warehouse architecture within the SQL Server environment and created ETL packages to efficiently load OLTP data into the Data Warehouse. Additionally, I utilized Power BI for data visualization, providing insightful and user-friendly representations of the data.
- Technologies used: **Dimensional Modeling,ETL,Data Visualization,Data Warehousing, Power BI**

Food Delivery Time Prediction System

2025

- Developed an end-to-end machine learning solution to accurately predict food delivery times using real-world delivery datasets. Performed extensive **data cleaning, feature engineering**, and implemented a complete preprocessing pipeline using **scikit-learn**. Built and trained a **Random Forest regression model** and optimized its performance using cross-validation and **hyperparameter tuning**.
- Technologies used: **Python, Pandas, NumPy, scikit-learn, matplotlib, seaborn, Random Forest, Feature Engineering, Data Visualization, Streamlit, Joblib**

SKILLS

- **AI / Machine Learning:** Python, NumPy, Pandas, Scikit-learn, TensorFlow, Keras, Deep Learning, NLP, LLMs, RAG Systems, Statistical Modeling, Random Forest, Transformers (RoBERTa, BERT, BART), Multi-Agent Systems
- **Generative AI & LLM Tools:** LangChain, LangGraph, LangSmith, Groq API, Google Gemini API, Prompt Engineering, MCP Servers, Sentence Transformers (BGE-M3), Vector Embeddings, Semantic Search, Hybrid Search
- **Data & Databases:** MySQL, PostgreSQL, MongoDB, NoSQL, JSONB, SSIS, SSMS, SSAS, SSRS, MSSQL, ETL, Dimensional Modeling
- **Data Visualization:** Power BI, Matplotlib, Seaborn, Chart.js, Streamlit
- **OCR & Document AI:** Tesseract, PaddleOCR, PPStructure, pdf2image, TF-IDF, Information Retrieval
- **Cloud & Big Data:** AWS, Azure, Hadoop, Spark
- **Web & Frameworks:** React JS, Node JS, Spring Boot, Express JS, Django, JavaScript, HTML, CSS, PHP
- **Programming Languages:** Python, Java, R, SQL, JavaScript, C, C++
- **DevOps & Tools:** Git/GitHub, Jupyter Notebook, Google Colab, VS Code, Postman, Android Studio, Excel, Joblib
- **UI & Design:** Figma, Android Studio, draw.io, Canva
- **IoT & Embedded Systems:** ESP32, MQTT, IoT Sensors, REST API

LANGUAGES

- Sinhala - Native
- English - Professional proficiency

ACHIEVEMENTS AND CERTIFICATES

- 2016/2020 works at Rahula College Board Of Prefects
- Installed as LEO in LEO Club of SLIIT
- AI/ML Engineer - Stage 1 - SLIIT (CODL)
- Python for Beginners – University of Moratuwa (CODL)

REFERENCES

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